Porch Lift for Nadine

A small bolt-together platform lift at the end of a 4 x 8 ft camper porch. 12 V, runs off the camper battery. Version 1.0, October 2026.

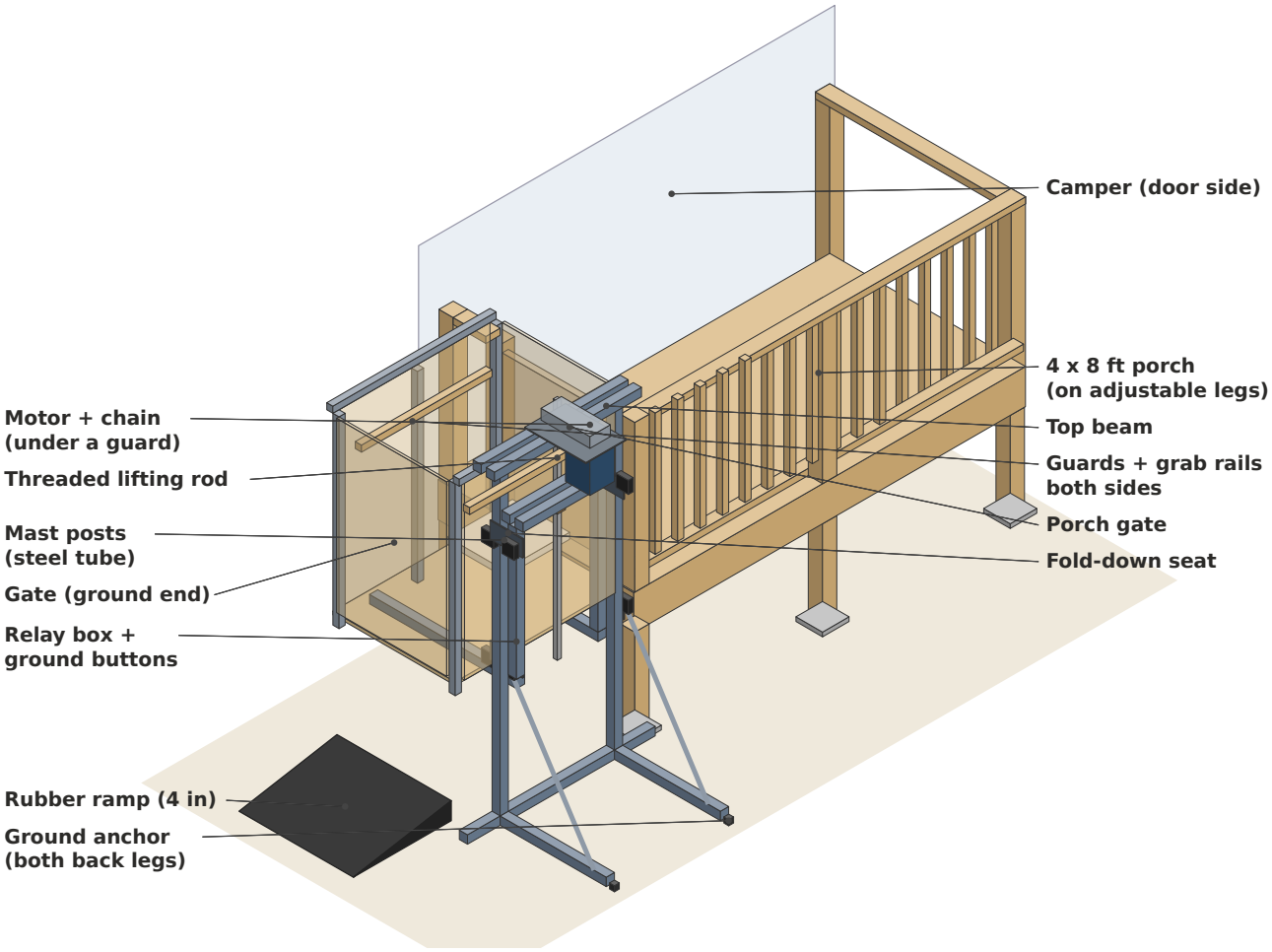


Figure 1: The finished lift at porch level. Nadine rolls her walker up the short ramp onto the platform, rides up, and walks straight off onto the porch.

What it does

- Lifts one standing or seated person (up to 300 lb, plus a walker) from the ground to the porch, 24 to 38 inches
- About 30 seconds per trip. Hold the button to move; let go and it stops
- Bolt-together steel: no welding. Comes apart in 3 pieces to move the camper

Why it's safe

- A threaded rod lifts it, not a cable. The thread can't slip, so if power fails it just stops and holds
- A second "backup" nut catches it if the main nut ever wears out
- Gates, a safety pan underneath, limit switches and 3 stop buttons
- Can be lowered by hand with a drill if the battery or motor dies

Please read first. This is a home-built lift, not a certified one. Build it carefully, follow every test in section 9 before anyone rides, and have a local welder or fabricator look over the finished frame. One rider at a time, 300 lb max. Never let children play on it. Don't use it for a wheelchair: the platform is only 30 inches wide.

1. How it works

A steel threaded rod (1 inch, ACME thread) hangs from the top of the mast. The motor turns it slowly through a bicycle-style chain. The platform hangs on a bronze nut that rides up and down that rod, like a nut on a bolt. The platform sticks out in front of the mast, and rollers on two steel posts keep it from tipping or swinging.

Dashed outline = carriage and deck at the top

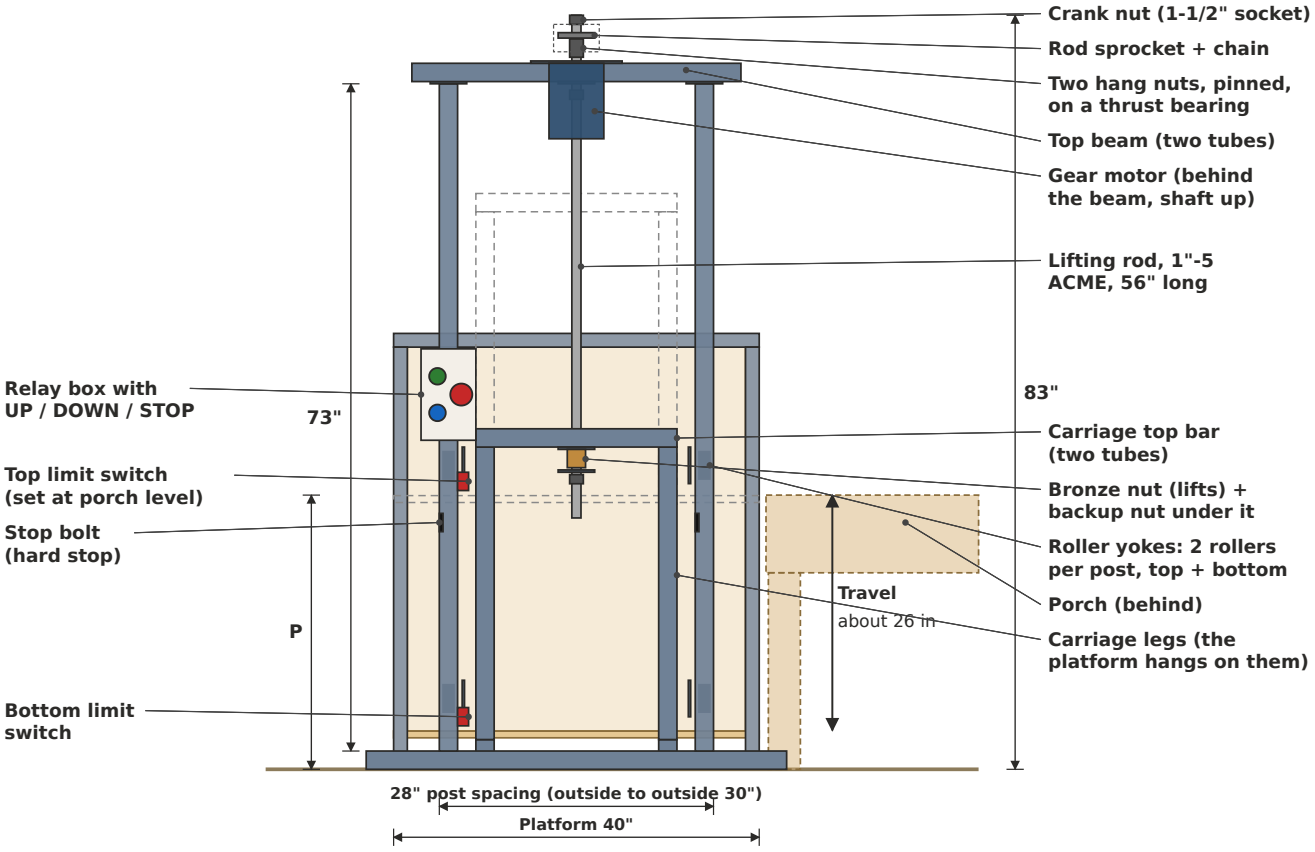
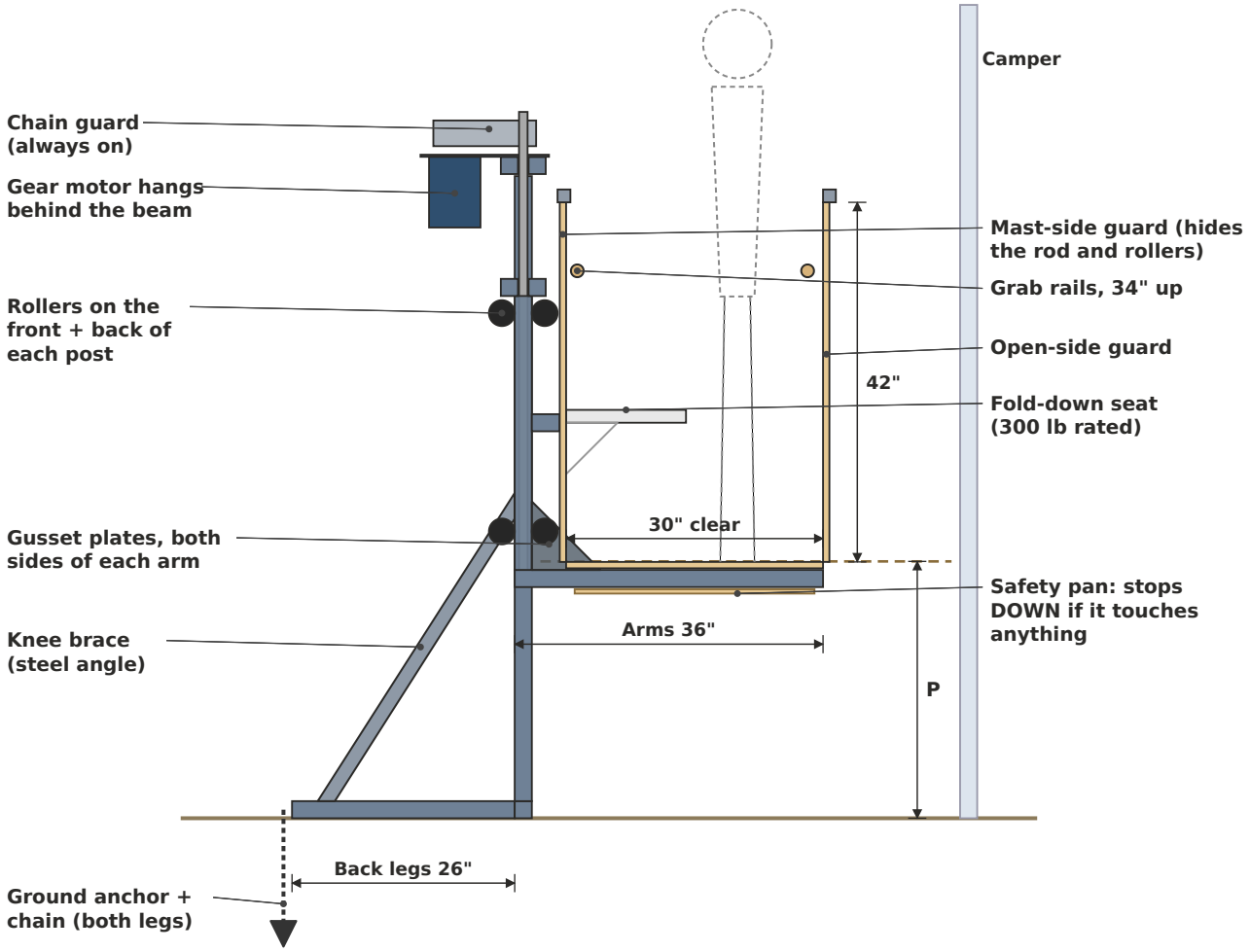


Figure 2: The mast, seen from the yard side (behind the lift). The carriage and platform are drawn at the bottom, and dashed at the top. P is your porch height.

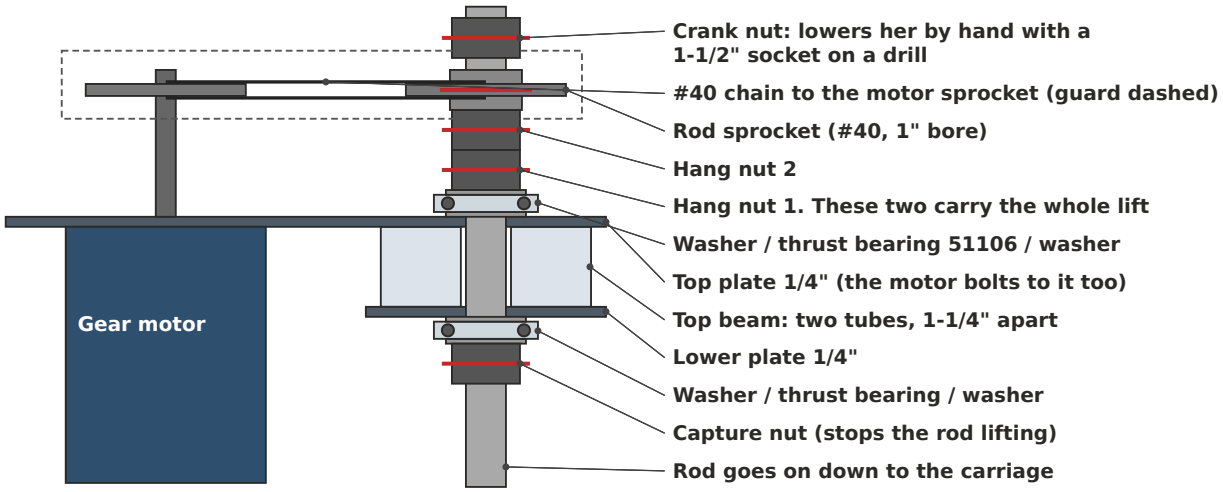


Seen from the porch end, platform at the top. Brown dashed line = porch deck level.

Figure 3: Side view from the porch end. The platform sticks out 30 inches in front of the mast. The ground anchors on the back legs keep the mast from tipping forward. Dashed person for scale.

Why a threaded rod and not a winch: this kind of thread (single-start ACME) is "self-locking". Weight on the platform can't turn the rod, so nothing can let it fall: not a dead battery, a broken wire, a broken chain or a failed motor. That's why it must be a plain single-start ACME rod: a multi-start rod or a ball screw would spin and drop.

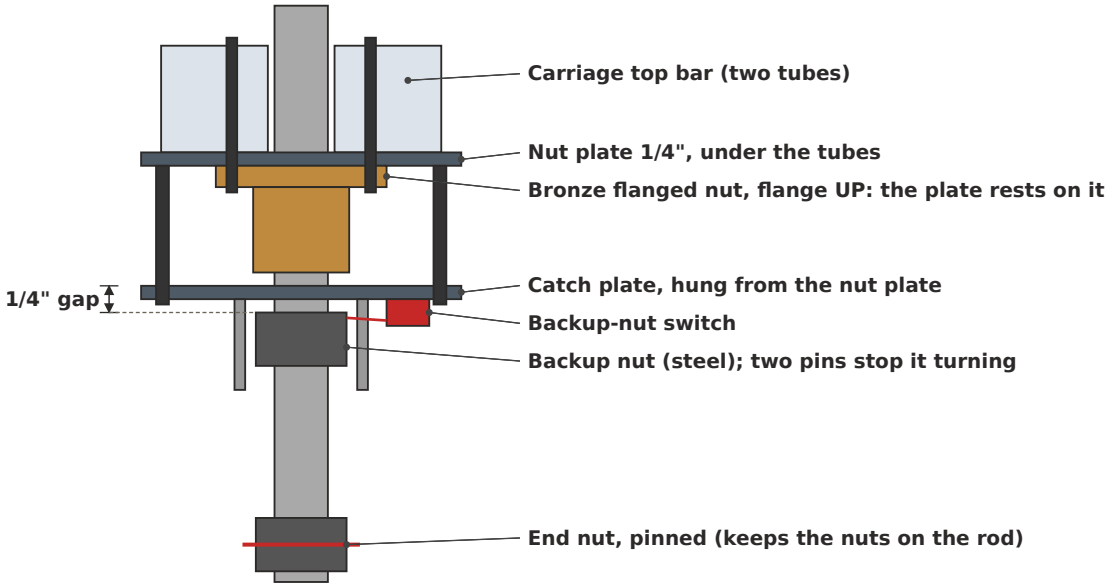
2. Detail drawings



— = 1/4" roll pin through the nut (or sprocket) and the rod, drilled by a machine shop

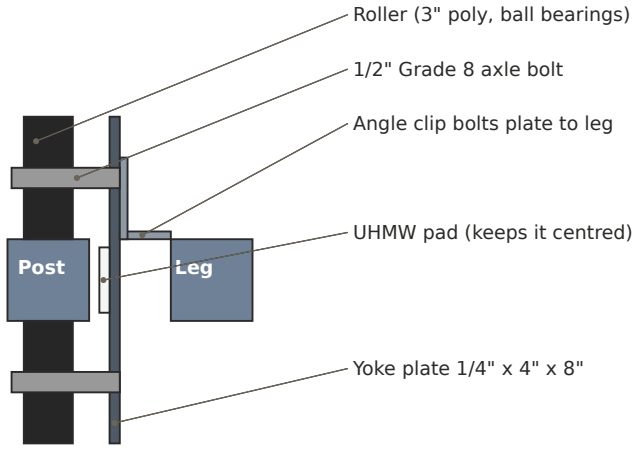
Not to scale. Leave about 1/32" of up-and-down play before drilling the capture nut.

Figure 4: Top of the mast, cut through the middle. The rod hangs from two pinned nuts sitting on a thrust bearing. A third pinned nut under the beam stops it lifting.



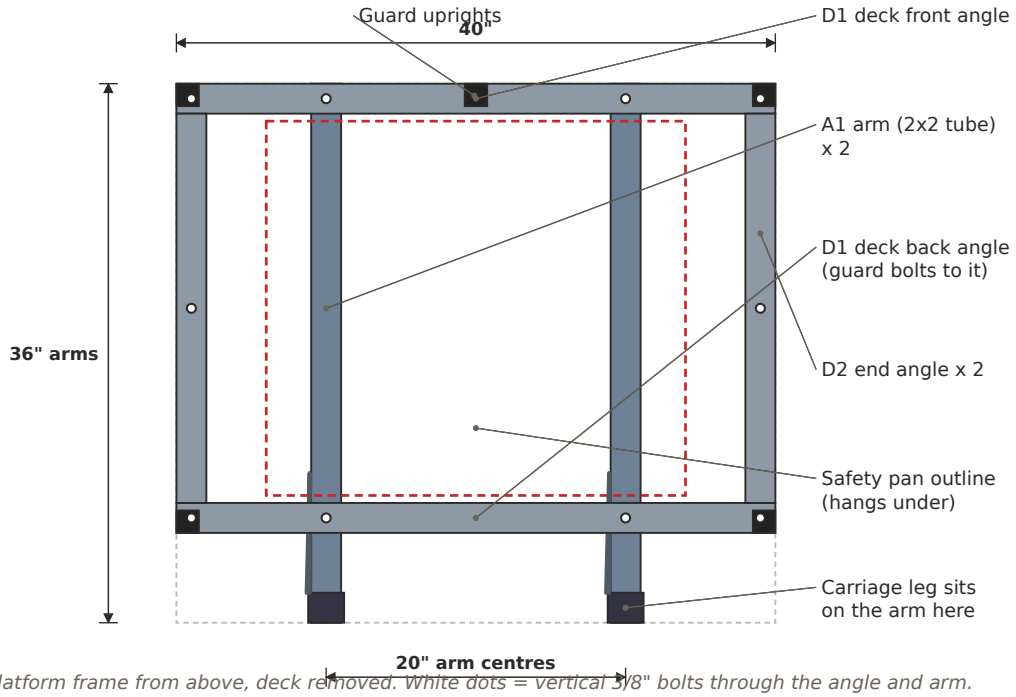
If the bronze nut's thread ever wears out, the carriage drops 1/4" onto the backup nut and stops there. The switch opens and the lift won't run until it's fixed.
Check the gap every month: if it's under 1/8", replace the bronze nut.

Figure 5: How the platform hangs on the rod, cut through the middle. The bronze nut lifts; the steel backup nut just follows along, 1/4 inch below, until it's needed.



Top view. 4 yokes (2 per post, top and bottom). The front roller rides the front face of the post, the back roller the back face. Mirror it on the other post.

Figure 6: Roller yoke. Four of these hold the carriage to the posts.



Platform frame from above, deck removed. White dots = vertical 3/8" bolts through the angle and arm.

Figure 7: Platform frame. The two arms carry everything; the angles hold the deck and guards.

3. Where it goes

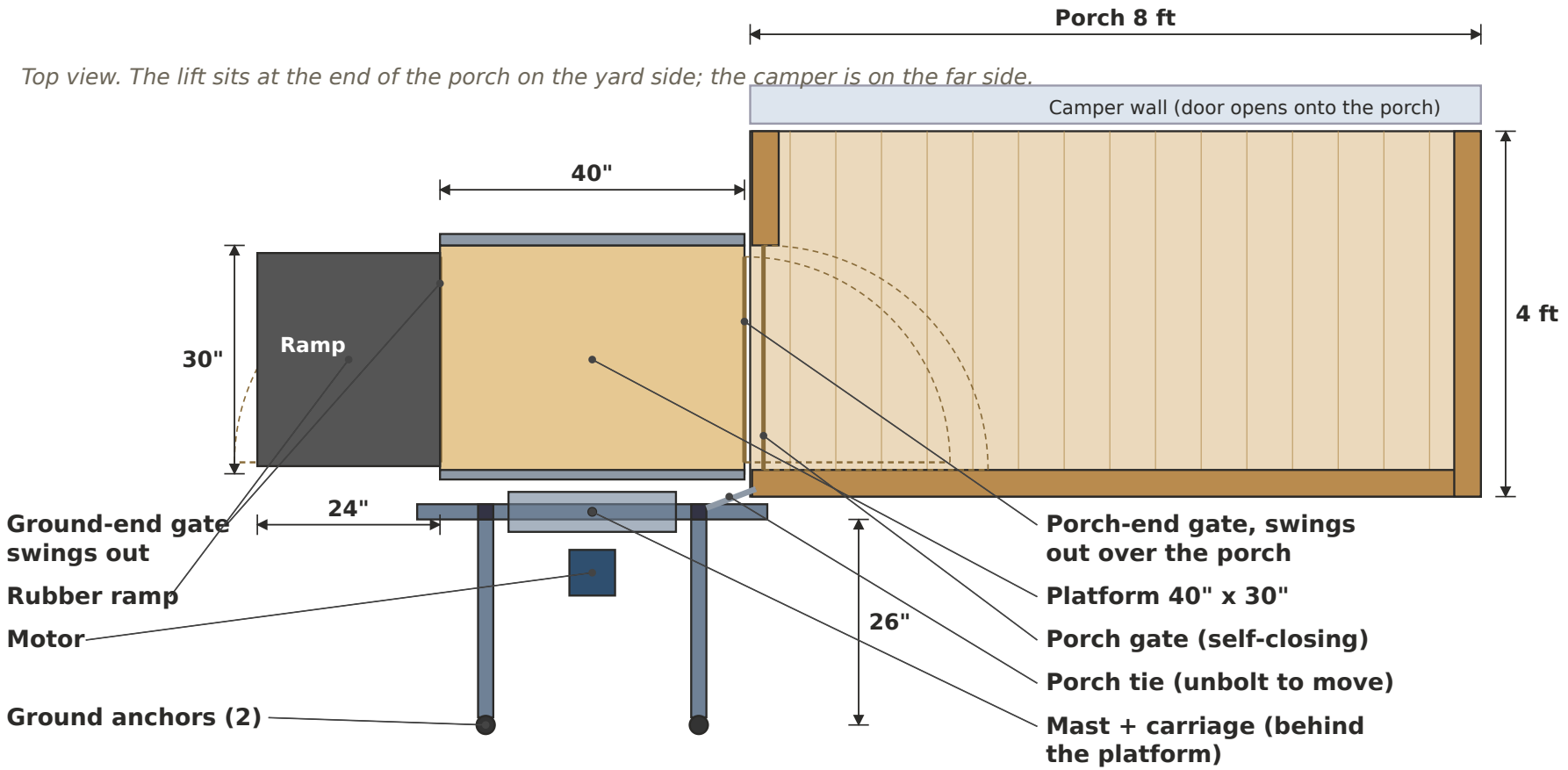
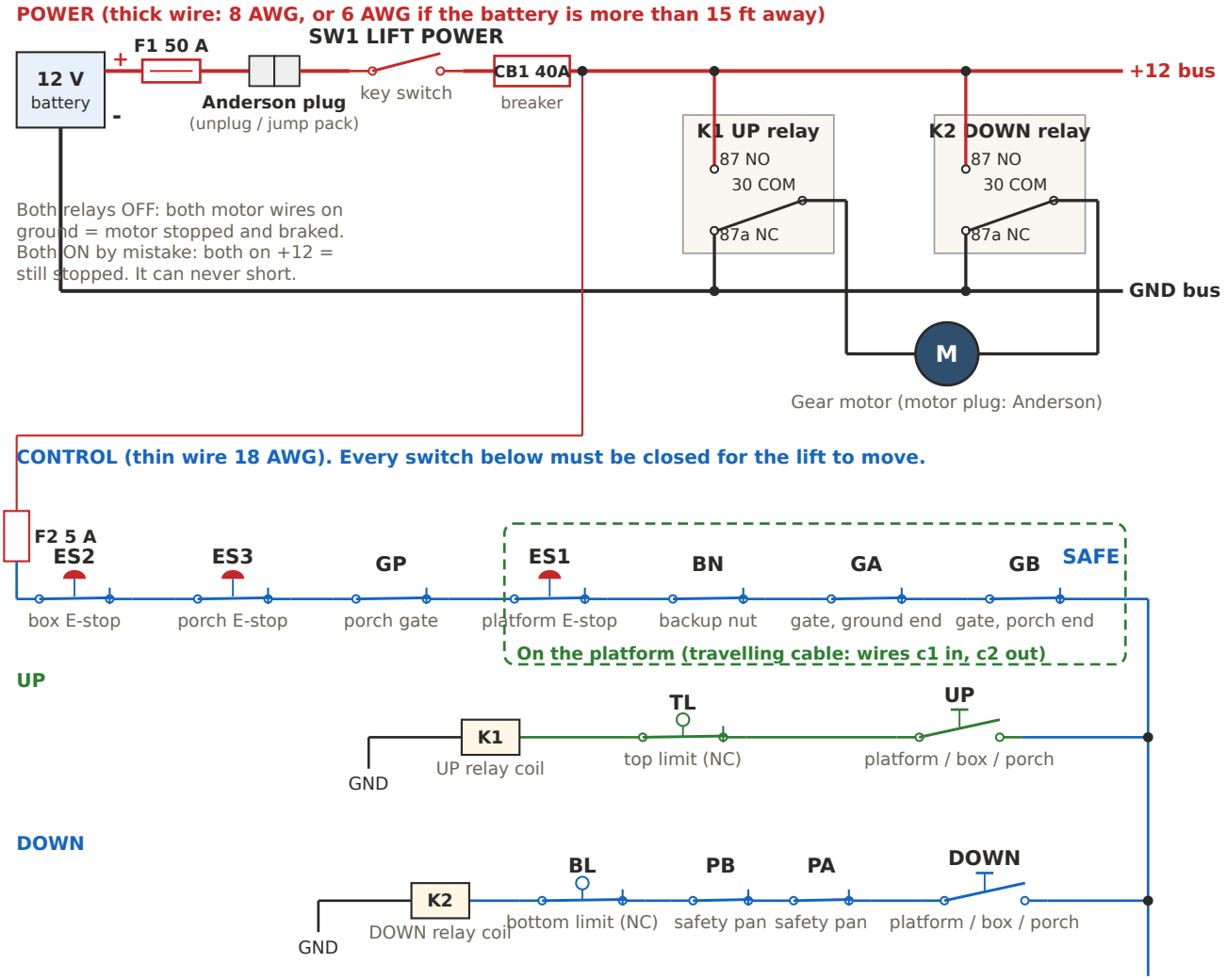


Figure 8: Top view of the porch and lift. Leave about 9 x 6 ft of firm, level ground for the lift, ramp and back legs.

4. Wiring



The three UP buttons (platform, relay box, porch) are wired side by side (in parallel); so are the three DOWN buttons. Hold to run: let go and it stops. Gate and pan switches are closed only when the gate is latched / the pan is free. Platform buttons, pan, gate and backup-nut switches ride on the platform: they connect through two 18/3 cords (6 wires) that hang in a loop from the relay box to the carriage. Use relays with a built-in coil diode.

Figure 9: Wiring diagram. No computer is in the safety chain: every switch simply breaks the circuit. Red = battery power, blue/green = control.

5. Cut lists

Give these lists to the steel supplier. Most will cut to length for a small fee. Lengths are for porch heights up to 38 inches.

2" x 2" square steel tube, 1/8" wall (11 gauge)

ID	Piece	Qty	Length	Notes
B1	Base rail	1	46"	Lies on the ground under the mast
B2	Back legs	2	26"	Stick out behind the mast; anchors at the ends
P1	Posts	2	73"	The rollers run on these: keep the faces clean and smooth
T1	Top beam tubes	2	36"	Side by side, 1-1/4" apart, rod between
A1	Platform arms	2	36"	Carry the whole platform
L1	Carriage legs	2	32"	Stand on the back ends of the arms
C1	Carriage top tubes	2	22"	Side by side, 1-1/4" apart, rod between
S1	Seat bar	1	22"	Across the fronts of the carriage legs, 18" above the deck
S2	Spacers	4	3"	Seat bar / leg tops to the mast-side guard

2" x 2" x 3/16" steel angle

ID	Piece	Qty	Length	Notes
K1	Knee braces	2	44"	Trim to fit: post side to back leg side
D1	Deck front + back	2	40"	Across the arms
D2	Deck ends	2	26"	Between the D1s at each end
CL	Clips	16	4"	4 post bases, 4 post tops, 4 yokes, 4 leg tops

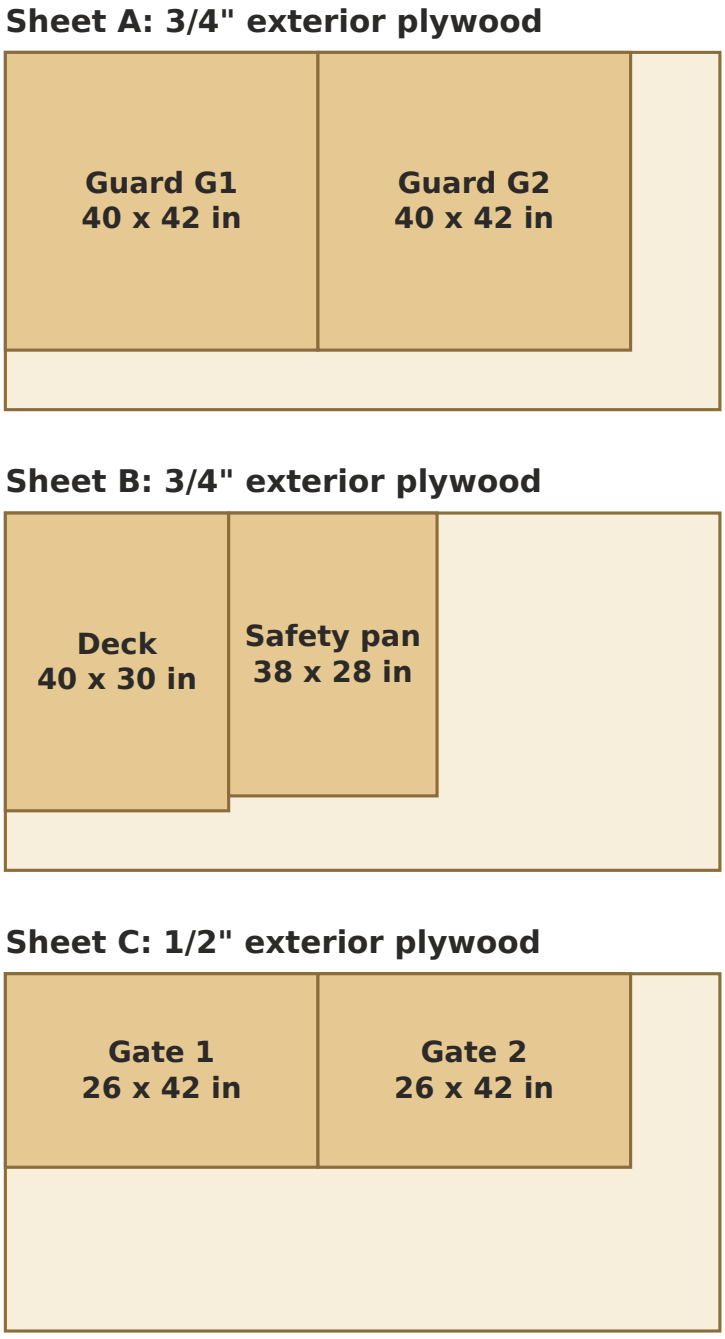
1-1/2" x 1-1/2" x 3/16" steel angle

ID	Piece	Qty	Length	Notes
U1	Guard uprights	5	46"	4 corners + middle of the open side
R1	Guard top rails	2	40"	Along the top of each guard
PT	Porch tie	1	24"	Trim to fit: right post to porch rim

Plate

Stock	Piece	Qty	Size
1/4" x 4" flat bar	Post caps (2), leg-top caps (2), lower plate (1), nut plate (1), catch plate (1)	7	6" long each
1/4" x 4" flat bar	Yoke plates	4	8" long each
1/4" plate	Top plate (motor + rod)	1	10" x 15"
1/4" plate	Arm gussets: right triangles	4	8" x 8" (cut 2 squares corner to corner)
1/4" plate	Base corner plates	2	6" x 6"

Holes: 3/8 inch bolts go in 25/64 inch holes; the 1/2 inch roller axles in 33/64 inch holes. The rod goes through 1-1/8 inch holes in the plates. Most holes are drilled at assembly: clamp the two pieces where they go and drill through both, so they always line up. A drill press or a magnetic drill makes this much easier.



Each sheet is 4 x 8 ft. The pale area is spare.

Figure 10: Plywood cutting plan.

6. Parts list

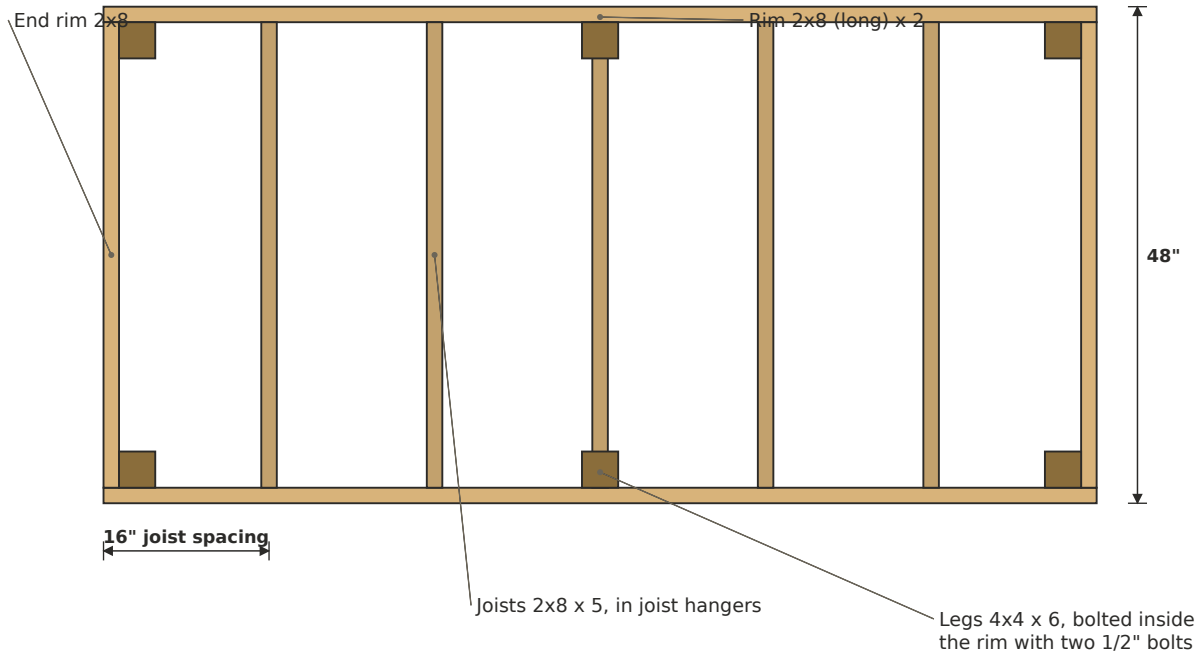
Prices are rough October 2026 numbers for new parts.

Group	Part	Qty	Rough cost
Steel	2"x2" square tube, 1/8" wall (11 ga)	44 ft (two 24 ft sticks)	$150-200
Steel	2"x2"x3/16" angle	24 ft	$60-80
Steel	1-1/2"x1-1/2"x3/16" angle	28 ft	$55-70
Steel	1/4"x4" flat bar 8 ft + one 24"x24" piece of 1/4" plate		$60-80
Drive	1"-5 ACME threaded rod, steel, right hand, single start (NOT multi-start, NOT a ball screw), 6 ft	1	$60-80
Drive	1"-5 ACME flanged bronze nut	1	$40-55
Drive	1"-5 ACME hex nut, steel	6	$50-70
Drive	Thrust ball bearing 51106 (30x47x11 mm) + 1" hardened washers	2 + 4	$20
Drive	12 V DC gear motor: 300-400 W, 300-400 rpm, 10 N.m or more, spur or planetary gearbox (NOT worm), keyed shaft	1	$90-130
Drive	#40 sprockets, 12 tooth: one 1" bore, one to fit the motor shaft; #40 chain 2 ft + master link		$45-55
Drive	1/4" x 1-1/2" roll pins (machine shop drills the holes)	6	$5 + shop time
Rollers	3" x 1-1/4" polyurethane wheels, ball bearings, 1/2" bore, rated 350 lb or more	8	$40-55
Rollers	1/4" UHMW plastic, 2" x 6" pads	4	$10
Bolts	3/8"-16 Grade 8 bolts (2-1/2" to 6-1/2"), nylock nuts, 2 washers each	about 90	$70-90
Bolts	1/2"-13 Grade 8 x 3-1/2" (roller axles) and x 3" (stop bolts), nylock nuts	8 + 2	$15
Wood	Exterior plywood: two 3/4" sheets + one 1/2" sheet	3	$130-160
Wood	1-1/2" round wooden handrail, 36" + 4 brackets	2	$35
Platform	Heavy self-closing spring hinges (4) + self-latching gate latches (2)		$50-60
Platform	Fold-up shower seat rated 300 lb or more	1	$45-60

Platform	Anti-slip deck paint or grip tape; primer + enamel for the steel		$40-50
Ground	4" rubber threshold ramp, 30" or wider	1	$40-50
Ground	30" auger ground anchors + 3/8" chain, turnbuckles, shackles	2 sets	$60-70
Electric	50 A fuse + holder, 40 A breaker, key switch, 5 A inline fuse		$45
Electric	40 A 12 V SPDT automotive relays with sockets (built-in coil diode)	2	$15
Electric	Red mushroom E-stop buttons (NC, twist to release)	3	$25
Electric	Weatherproof momentary push buttons (UP + DOWN at 3 stations)	6	$20
Electric	Heavy-duty roller-lever limit switches, NC	2	$20
Electric	Sealed roller-lever micro switches (3 gates, 2 pan, 1 backup nut)	6	$20
Electric	8 AWG red + black cable (length to the battery), 18/3 SJOOW cord 2 x 10 ft, 18/4 cord to porch station		$70-90
Electric	Anderson SB50 connectors (2 pairs), weatherproof boxes (3), terminals, loom		$45

> **Cost:** the lift comes to about **$1,450-1,800** in new parts, plus machine-shop time, and the porch adds about **$350-450**. That's over the $1,000 goal. Ways to bring it down: buy the steel at a scrap or surplus yard by the pound (often half price), use plywood and lumber from Facebook Marketplace, and skip the porch call station. Also look for a **used commercial porch lift**: they sometimes sell for $1,000-2,000 and are already certified.

7. Porch (4 x 8 ft)



Lift end is on the left (x = 0). Deck boards run across the 48" width. Add 2x4 knee braces from each leg to the rim.

Figure 11: Porch framing from above. Legs bolt to the frame, so the porch comes apart for moving.

Lumber	Qty	Use
2x8 treated, 8 ft	6	Long rims (2 full length); cut the rest into 7 pieces 45" (2 end rims + 5 joists)
4x4 treated, 8 ft	5	6 legs (length P minus 9", check on site) + 6 railing posts 44"
2x4 treated, 8 ft	7	Railing top + bottom rails, porch gate frame, 6 knee braces 30"
2x2 balusters, 36"	36	4" gaps or less between them (yard side, far end, gate)
5/4 x 6 deck boards, 8 ft	9	Cut in half: 18 boards 48" across the frame
Concrete deck blocks	6	One under each leg, on firm level ground
2x8 joist hangers, 1/2" x 6" carriage bolts (12), deck screws, structural screws		
Self-closing gate hinges (2) + latch (1) for the porch gate		Its latch also holds the GP switch closed

- **Height:** make the top of the deck level with the camper door threshold, or one step below it. Your porch is **30 inches** high (P = 30), which is what all the

drawings use. (The design works from 24 to 38 inches.)

- **Railings:** 36 inches high on the yard side and far end, with a 33 inch self-closing gate at the lift end. Leave the camper side open to the door.
- **Lift end:** face the end rim with smooth plywood, and keep the gap to the platform at 3/4 inch or less so a toe can't catch.

8. Build steps

1. **Plan.** Your porch height P is 30 inches. Pick firm, level ground at the end of the porch. Call 811 (free) before screwing in ground anchors, to check for buried lines.
2. **Buy the motor first.** Then order the motor sprocket to fit its shaft. Order the steel cut to the lists in section 5.
3. **Machine shop.** Take the rod, the 6 steel nuts and the rod sprocket to a machine shop. Have them cut the rod to 56 inches. First thread everything onto the rod in the order of Figure 4 with the plates clamped between, mark each nut, then have them drill a 1/4 inch hole through each nut and the rod and tap in a roll pin. Do the same for the end nut (Figure 5). **Don't drill the capture nut yet** (step 9).
4. **Paint.** Deburr every cut and hole, then prime and paint all the steel. Don't paint the rod or the faces of the posts where the rollers run.
5. **Base.** Lay the base rail B1 on level ground. Bolt the back legs B2 to it with a corner plate on top and a clip inside each joint. Check it's square: measure the two diagonals; they should match.
6. **Posts.** Stand the posts P1 on the base rail, 28 inches apart centre to centre, with a clip on the front and back of each. Bolt the knee braces K1 from each post (about 38 inches up) to its back leg. Check both posts are plumb both ways with a level, and brace them temporarily.
7. **Top beam.** Bolt a cap plate on top of each post with two clips. Set the two beam tubes T1 on the caps with a 1-1/4 inch wood block between them, and bolt them down through the caps. Bolt the lower plate under the middle and the top plate on top, and drill the 1-1/8 inch rod hole through both. Hang a plumb bob through the hole: the rod line must be straight up and down.
8. **Carriage.** On the ground, bolt each carriage leg L1 on the back end of an arm A1, with a gusset plate on each side (3 bolts into the leg, 3 into the arm). Bolt the top tubes C1 across the leg tops with cap plates, leaving a 1-1/4 inch gap. Bolt the nut plate under them and the bronze nut under that, flange up (Figure 5). Hang the catch plate below. Bolt on the four roller yokes, but leave the rollers off for now.

9. **Hang the carriage.** Set the carriage on blocks under the mast with the arms pointing away from it. From the top, feed the rod down through the plates and beam, then screw it into the bronze nut by turning the rod. Add the backup nut and the end nut. Fit the top stack: washer, bearing, washer, hang nuts and sprocket above; washer, bearing, washer, capture nut below. Set about 1/32 inch of end play, then mark and drill the capture nut. Now fit the rollers: each should just touch its post face. Set the backup nut 1/4 inch below the catch plate and fit its anti-turn pins.
10. **Deck and guards.** Bolt the deck angles D1 and D2 to the arms (Figure 7). Bolt down the plywood deck and paint it with anti-slip paint. Bolt the 5 uprights U1, then the two guards, top rails R1, gates, grab rails at 34 inches, and the seat on the seat bar S1 with the S2 spacers.
11. **Motor and chain.** Bolt the motor under the top plate in its slotted holes. Line up the two sprockets with a straightedge and set about 1/4 inch of chain slack. Fit the chain guard. With the motor unplugged, turn the crank nut with a drill: the platform should run smoothly all the way up and down. Fit the stop bolts through the posts 1 inch above the top yokes at full height.
12. **Switches.** Hang the safety pan 1/2 inch under the deck on 4 bolts in slotted holes, so it lifts if anything pushes it up, and fit the two pan switches. Fit the gate switches (closed only when the gate is latched) and the backup-nut switch. Clamp the limit switches to a post: bottom, so it stops with the pan 1/2 inch off the ground; top, so the deck stops level with the porch.
13. **Wiring.** Wire it as Figure 9. Mount the relay box on the ground-end post with the ground buttons and the E-stop. Run the two 18/3 cords from the box to the carriage in a hanging loop long enough for full travel, clamped at both ends. Run the power cable from the battery with the 50 A fuse within 7 inches of the battery.
14. **Set it up.** Build the porch (section 7). Put the lift at its end with a 3/4 inch gap. Screw in the ground anchors behind the back legs and tighten the chains with the turnbuckles. Bolt the porch tie PT from the right post to the porch rim. Put the rubber ramp at the ground-end gate.

9. Testing (before anyone rides)

1. **Each switch, empty.** Hold UP, then press each of these in turn: every one must stop it: the 3 E-stops, both gates, the porch gate, the backup-nut switch, the top limit. Do the same going DOWN with the pan (push it up by hand) and the bottom limit. Check all three button stations.

2. **Load test.** Put sandbags on the platform at 1.5 times the heaviest rider (450 lb for a 300 lb rider), spread out. Run it up and down 20 times. Watch the battery current with a clamp meter: it should stay under 30 A. Afterwards, check the backup-nut gap is still 1/4 inch and every bolt is still tight.
3. **Hold test.** Leave the loaded platform at the top for 10 minutes. It must not creep down at all.
4. **Hand-lowering test.** With the load still on, unplug the motor and lower it with the drill on the crank nut.
5. **First ride.** A helper rides first, then Nadine with someone standing by. Have her try the seat too.

10. Upkeep

- **Every month:** test every switch (step 9.1). Check the backup-nut gap: **if it's under 1/8 inch, stop using the lift and replace the bronze nut.** Wipe and grease the rod with lithium grease. Check the chain and the anchors.
- **Every 6 months:** check every bolt is tight, and look for rust and cracks.
- **If the battery dies with her on it:** plug a jump pack into the Anderson plug, or unplug the motor and lower it with a drill and a 1-1/2 inch socket on the crank nut. That's about 150 turns, around a minute with a drill.
- **Moving the camper:** lower the platform to the ground, unplug it, unbolt the porch tie and remove the anchors. It comes apart in 3 pieces: base, mast and carriage. The carriage weighs about 250 lb, so take the guards and gates off first and have 3 people lift it.

> **Home Assistant (optional, later):** a small ESP32 can watch the lift and send alerts such as "lift in use", "stopped by a safety switch" and low battery. It would only watch: someone has to be there whenever it moves, so it should never run it from a phone.